Part Ⅰ. Data Collecting, Processing, and Plotting (50p).

The assignment will test the student's ability to read and process datasets, visualize data with Python and Matplotlib, analyze data using visualization techniques, and select an appropriate data representation based on the type of data and/or the question being asked.

Case A: Stock Performance Analysis

Imagine you're a financial analyst working for an investment company. The company is interested in tracking the performance of a portfolio that includes some of the world's largest and most influential companies across sectors such as technology, finance, and retail. Your analysis will involve multiple critical areas: stock performance, correlation analysis, and comparisons across different periods.

Your team has provided you with a small portfolio (i.e., dataset) for early-stage assessment. The dataset you are working with includes the **daily closing prices (DCPs)** of 5 major companies (AAPL, MSFT, GOOGLE, AMZN, TSLA) in USD over 100 trading days. However, as multiple people manually retrieve the data from different sources, the dataset is poorly sorted and may contain missing values.

CT1. Data Exploration and Static Visualization (25p)

CT1-1: Explore the dataset
Before starting to analyze the dataset, you aim to familiarize yourself with the dataset and perform an essential exploration to understand its structure, content, and the nature of the data you're working with. This will set the foundation for more in-depth analyses in subsequent tasks.

(1) Read the dataset ("assignment_CaseA_data.csv") in a DataFrame and display the first 5 rows and last 10 rows of the dataset.
(2) Check whether the dataset contains missing values and finds those affected data samples in your code. For later analysis, these data samples ("rows") should be processed. Write text reports on your methods of processing any missing values.
(3) Sort the dataset by date from recent to earliest in your code and write text answers of the time span of the dataset.

CT1-2: Comparison and relationship
Based on your initial exploration from Task 1, this task focuses on comparing the stock prices of different companies and examining relationships between them. You will use visualizations to gain insights into how these stocks behave relative to one another.

(1) Visualize the observed DCPs for the company AMZN and find the date when the DCP reaches its lowest point and highlight it in the plot.
(2) Your task is to compare the DCPs (or stock price movements) of the five companies over time. Which is the most appropriate chart in this case? Plot the data using the chosen chart and identify which companies have similar or divergent trends. Identify which companies have similar or divergent trends in text answers.
(3) You now have to focus on the data of the company AAPL. Select and plot the most appropriate chart for the following task: identify the price intervals having the highest frequency (i.e., the most common price intervals). The price range can be divided into 10-USD intervals.

Case B: Employee Productivity Analysis (12.5p)

Most companies use different performance metrics to assess employee productivity. *R&D, Marketing*, and *Sales*).

Task 1: Correlation of performance factors
Factors that influence the productivity of an employee are often related to each other to some degree. Your goal is to determine how the three factors (*hours worked per month, tasks completed,* and *efficiency score*) are correlated among employees across departments. The dataset is provided ("assignment_CaseB_performance.csv").

Calculate the correlation (using corr() method) between the three performance factors mentioned above across all employees and visualize the correlation matrix using a heatmap to represent the strength and direction of the correlation between the three factors. Then analyze**:**

- Do employees who work more hours generally complete more tasks?
- Which factors have the strongest positive or negative correlation, and what might this indicate about employee productivity in each department?

Requirements:

(1) Use Matplotlib to create the heatmap.
(2) Choose an appropriate colormap and justify the selection.
(3) Ensure the visualization is clear, with axis labels and a color bar that indicates the correlation strength.

Task 2: Identify employees' performance
Supposing that the company has observed 10 employees who contributed good performance in the *Sales* department based on a performance rating with 5 factors: *Hours Worked Per Month, Tasks Completed, Efficiency Score, Attendance Rate,* and *Customer Satisfaction*. The company wants to assess which employees are the most balanced performers in these five dimensions. Some employees may be stronger in specific areas, while others may perform more consistently across all metrics.

Plot a radar chart that shows the performance of all 10 employees in one figure. Then analyze:

- Which two employees are well-rounded across all dimensions, and which one excels in customer satisfaction?
- Identify any employees with noticeable strengths or weaknesses based on the radar chart.

Requirements:

(1) Each employee should have a unique line in the radar chart, and the axes should represent the five dimensions.
(2) Ensure the data is normalized so that the values of each metric can be compared easily.
(3) Add a legend to help distinguish between the employees.
(4) Design the chart trying to keep high readability, e.g., minimize the overlapping and maximize the distinctness of the visual encoding for each employee. Describe your design choices briefly in text answers.